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Perfluoroalkyl compound levels in breast milk of mothers in Riyadh, Kingdom of Saudi Arabia

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Per- and Polyfluoroalkyl substances (PFAS) are persistent environmental contaminants widely used in consumer and industrial applications including stain- and water-repellent textiles, nonstick cookware, firefighting foams, and food packaging. Their persistence and bioaccumulation potential raise concerns about early-life exposure through breastfeeding. Despite increasing international attention, no biomonitoring data on PFAS in breast milk have been reported from Saudi Arabia. The study aim to quantify perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) in breast milk from lactating women in Riyadh and estimate infant exposure via breastfeeding using standardized exposure assessment methods. Breast milk samples were collected from 25 mothers within 30 days postpartum. Samples were analyzed by LC–MS/MS following solid-phase extraction. Estimated daily intake (EDI) for PFOA and PFOS was calculated using the equation EDI = (CBM × VBM) / BW, assuming an average infant body weight of 2.5 kg and daily milk intake of 292.5 mL during early infancy. Quantifiable concentrations (> LLOQ = 25 ng/L) were detected in 4 of 25 participants. PFOA concentrations ranged from < LLOQ to 73.3 ng/L, and PFOS concentrations ranged from < LLOQ to 85.2 ng/L. Median concentrations for both compounds remained < LLOQ. Infant EDI values were below the U.S. EPA oral reference dose (20 ng/kg-bw/day) but exceeded EFSA benchmark values in 12% of samples for PFOA (EFSA = 0.857 ng/kg-bw/day) and 16% of samples for PFOS (EFSA = 1.857 ng/kg-bw/day). This study provides the first biomonitoring evidence of PFAS in breast milk in Saudi Arabia, demonstrating low but measurable exposure among lactating women. The findings underscore a need for expanded biomonitoring and assessment of potential exposure sources in the region.

Keywords PFAS, Child, Breast milk, Breastfeeding, LC-MS/MS, Regulations, Biomonitoring, PFOA, PFOS, Environmental pollutants

Per- and polyfluoroalkyl substances (PFAS) are a broad class of synthetic, fluorinated organic chemicals used since the mid-20th century in industrial processes and consumer products including surfactants, stain- and water-repellent textiles, firefighting foams, food packaging, and cookware[1–5]. PFAS are structurally characterized by carbon–fluorine bonds that confer high stability and extreme resistance to environmental degradation, leading to global distribution and biological persistence[1]. Among these compounds, perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) are the most widely studied due to their bioaccumulative behavior, and documented human health concerns[6–14]. Consequently, these compounds were listed under the Stockholm Convention on Persistent Organic Pollutants, and major manufacturers committed to phase-out programs such as the U.S. PFOA Stewardship Program[15]. Exposure occurs through multiple pathways including contaminated drinking water, food, soil, household dust, and consumer products[16–20]. Several human biomonitoring studies have measured PFOA and PFOS in serum, urine, breast milk, and cord blood, providing evidence of both prenatal transplacental transfer and postnatal transfer through breastfeeding. Early-life exposure has gained increasing attention due to heightened biological vulnerability during fetal and infant development[21–26]. Studies have reported associations between PFAS exposure and altered immune function (including reduced vaccine antibody response), endocrine disruption, neurodevelopmental effects, lipid metabolism changes, and reduced birth weight[21–26]. Breast milk PFOA and PFOS concentrations have been reported in numerous regions. However, geographic disparities exist, and data remain limited for many countries in the Middle East.

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To date, no published study has examined PFOA and PFOS levels in breast milk from Saudi Arabia, despite rapid industrial and urban expansion and increasing concern regarding potential environmental contamination. Understanding local exposure levels is essential for contextualizing infant risk and comparing findings with international data. Therefore, the present study aimed to quantify PFOA and PFOS concentrations in breast milk from Saudi mothers residing in Riyadh, estimate infants' daily intake (EDI) of these compounds via breastfeeding, and compare observed levels with previously published literature and relevant exposure screening values. While several studies have reported associations between PFAS exposure and adverse health outcomes, the current study does not evaluate health endpoints. Accordingly, findings should be interpreted as exposure characterization rather than evidence of health impact.

Materials and methods

All experimental procedures were conducted with KAIMRC IRB guidelines. The study received IRB approval under the IRB approved project number (RC21R/075/03), Informed consent was obtained from subjects. Blank consent form is available for review upon request.

The research was performed in accordance with the Declaration of Helsinki ethical principles for medical research which involves human subjects.

Reagents and chemicals

The analysis of PFOA and PFOS typically involves several steps to isolate these compounds from complex matrices like water and milk. Thus, various reagents and chemicals were required for the sample preparation. Labeled internal standards MPFOA and MPFOS were also essential for the accurate determination of PFOA and PFOS concentrations. All information about standards, chemicals and materials used in this study is presented in Table 1.

Study population and sample collection

This study was designed as an exploratory biomonitoring assessment and was not powered for inferential statistical comparisons. A sample size of 25 participants was considered adequate for the primary objective of characterizing presence and concentration ranges of PFOA and PFOS in breast milk within a previously unstudied population. Similar exploratory sample sizes have been used in breast milk PFAS biomonitoring studies, particularly when analytical costs and recruitment logistics constrain large cohort enrollment. Findings should therefore be interpreted as preliminary and hypothesis-generating[27]. Breast milk samples were collected in January 2024 at King Abdullah Specialist Children's Hospital in Riyadh. Eligible participants were breastfeeding mothers who had delivered within the past 30 days. Mothers who had stopped breastfeeding, delivered more than one month prior, or declined participation were excluded. The hospital serves a large and diverse population, facilitating access to eligible participants. After obtaining written informed consent, each participant received a sterile, single-use breast milk collection kit containing a disposable manual breast pump. Breast milk was expressed directly into sterile polypropylene tubes, immediately labeled, and transported under refrigerated conditions. All samples were placed in a − 20 °C freezer within three hours of collection. Maternal age and pregnancy type (primiparous vs. multiparous) were obtained at the time of sample collection through brief interviewer-administered questionsand are summarized in Table 2. Participants' ages ranged from 25 to 42 years (mean ± SD: 32 ± 4 years), and most breastfeeding mothers were multiparous (68%). Other potential determinants of PFOA and PFOS exposure—such as maternal BMI, dietary patterns, residential water source, and breastfeeding duration—were not collected or assessed, as the study was designed as an exploratory exposure assessment rather than an investigation of exposure predictors.

Chemicals and reagents	Abbreviation	Molecular weight (g/mol)	Description	Supplier
Perfluoro-n-octanoic acid	PFOA	414.7	50 µg/mL, in 1.2 mL vial of methanol	Wellington Laboratories Inc., Canada
Sodium Perfluoro-1- octanesulfonate	L-PFOS	538.22		
sodium perfluoro-1-[1,2,3,4-13C4] octanesulfonate	MPFOS	526.08		
perfluoro-n-[1,2,3,4-13C4] octanoic acid	MPFOA	418		
Methanol	MeOH	32.04	LC-MS grade	Honeywell
Acetone	C_3H_6O	58.08	HPLC grade	Sigma-Aldrich
Formic Acid	HCO_2H	46.03	HPLC grade	
Acetic Acid	AcOH	60.05	HPLC grade	Fluka
Ammonium acetate	$NH_4CH_3CO_2$	77.08	HPLC grade	
ammonium hydroxide	NH_4OH	35.04	~ 25% NH3 basis	
Supelclean ENVI-Cars	NA		500 mg SPE Tube	Supelco
Oasis HLB cartridge			500 mg SPE cartridge	Waters

Table 1. Details regarding the standards, chemicals, and materials utilized in the current study.

Age group (years)	*n*	(%)
20–29	10	40
30–39	13	52
40–49	2	8
Types of pregnancy		
Primiparous	8	32
Multipara	17	68

Table 2. Summary of demographic characteristics of study participants ($n = 25$).

Compound name	Parent (m/z)	Daughter (m/z)	Cone (V)	Collision (eV)	Retention time
PFOA	412.9000	169.0000	12	16	3.13
MPFOA	416.9000	372.0000	16	10	3.13
PFOS	498.9000	79.9000	58	36	3.21
MPFOS	502.9000	79.9000	52	36	3.21

Table 3. Mass transitions and parameters of the chemical compounds (native and internal standards).

Samples Preparation

Samples were prepared according to the extraction procedure described by Kadar et al. (2011) with slight modifications[28]. The workflow consisted of three main stages: (i) extraction from breast milk, (ii) solid-phase purification, and (iii) final concentration and reconstitution for LC–MS/MS analysis.

Extraction from breast milk

The samples were placed at room temperature to be thawed, and then 3 mL of milk sample was mixed with 100 µL of 0.02 ug/ml MPFOA and MPFOS as an internal standard, along with 9 ml of acetone. The samples underwent ultra-sonication for 30 s, followed by centrifugation at 20,000 rpm for 10 min. The samples were evaporated using a gentle stream of dinitrogen gas at 45 °C until the volume reached 3 ml. After this, the pH adjustment was performed by adding 8 ml of 0.1 M formic acid solution and thoroughly mixing it with the samples.

Solid-phase extraction and purification

Solid-phase extraction was conducted in two steps using an Oasis HLB 500 mg SPE cartridge followed by a Supelclean™ ENVI-Carb™ cartridge. The initial purification step on the Oasis® HLB cartridge began by conditioning the cartridge with 10 ml of methanol and then 10 ml of 0.1 M formic acid to ensure the solid phase remained wet. Following this, the sample was loaded, and after complete passage through the column, two washes were conducted using 5 ml of 0.1 M formic acid and then 5 ml of 0.1 M formic acid/methanol (50/50, v/v) solution. Vacuum was then applied for 5 min to remove any remaining drops. The analytes were subsequently eluted using 6 ml of a mixture of methanol and ammonia solution (33% ammonia, 99/1, v/v), then the eluate was concentrated to around 2 ml under nitrogen. The second purification step involved the Supelclean™ ENVI-Carb™ cartridge after being conditioned with 10 ml of methanol. The sample was loaded into the column and then eluted with 6 ml of a methanol/glacial acetic acid solution (80/1, v/v).

Final concentration and reconstitution

The eluate was evaporated entirely under nitrogen and reconstituted in 200 µL of methanol/water (50/50, v/v) solution. The resulting mixture was transferred to a 1.5 ml Eppendorf tube and centrifuged at 12,000 rpm (report the gravity instead) for 45 min. The supernatant (150 µL) was collected and combined with 50 µL of water in a glass vial for analysis using the LC–MS/MS system.

LC-MS/MS analysis

PFAS compounds (PFOA and PFOS) in breast milk samples were identified and quantified using liquid chromatography-tandem mass spectrometry (LC-MS/MS). The instrument consists of Xevo TQ-S triple-quadrupole mass spectrometer coupled with an electrospray ionization (ESI) source and ACQUITY UPLC I-CLASS system with a binary solvent manager that provides consistent solvent delivery (Waters Corporation, Milford, MA, USA). An ACQUITY UPLC BEH C18 column (2.1 × 50 mm, 1.7 µm, Waters, Milford, MA, USA) was used for chromatographic separation at a 0.300mL/min flow rate. The column temperature was set to 45 °C.

Two mobile phases were designated as mobile phase A and B. Mobile phase A contained 2 mM ammonium acetate in methanol with water (5%/95% v/v), while mobile phase B consisted of 2 mM ammonium acetate in methanol. Analysis was performed using a gradient elution program as follows: 0–0.50 min, 5% B; 0.50–2.50 min, 5–98% B; 2.50–5.00 min, 98% B; 5.00–5.10 min 98–5% B; 5.10–7.00 min, 5% B. The flow rate was at 0.300 mL/min. The injection volume was 5.000 µL, and the total run time was 7.0 min. PFAS compounds were identified using MRM (multi-reaction monitoring). The mass transitions and parameters of the selected compounds are shown in Table 3.

The parameters were set as follows: a Capillary voltage of 0.50 kV, cone voltage of 30 V, dissolution temperatures 55 °C, desolvation of gas flow 500 L/h, cone gas flow of 150 L/h and collision gas flow of 0.18 mL/Min. Data were processed using MassLynx software (Waters Corporation, Milford, MA, USA). The typical MRM chromatograms for PFOA, PFOS, and their isotopically labeled compounds are shown in Fig. 1.

Method validation of LC-MS/MS analysis

The analytical method was validated in accordance with internationally accepted bioanalytical guidelines for quantitative LC–MS/MS analysis. Method specificity was evaluated by analyzing Milli-Q water to confirm the absence of interfering signals and background contamination at the retention times of PFOA and PFOS. Quantification was performed using isotopically labeled internal standards to correct for instrumental variability and potential matrix effects.Calibration curves for PFOA and PFOS were constructed using seven concentration levels. For PFOA, calibration standards ranged from 25 to 1500 ng/L, while PFOS calibration ranged from 25 to 500 ng/L. Linear regression with 1/x weighting was applied using internal standard normalization, and the origin was excluded from the regression model. Excellent linearity was achieved for both analytes across all analytical runs, with correlation coefficients (r^2) of approximately 0.9987 for PFOA and 0.9933 for PFOS as shown in Fig. 2. Calibration accuracy for individual concentration levels ranged between approximately 92% and 110%, meeting accepted performance criteria for quantitative bioanalytical methods.The limit of quantification (LOQ, reported as LLOQ) for both PFOA and PFOS was established at 25 ng/L, corresponding to the lowest validated calibration level demonstrating acceptable accuracy and precision. Although detectable signals were observed below 25 ng/L, the method was not validated for quantitative determination below the LOQ; therefore, concentrations below this level were not reported as numerical values and were classified as < LOQ (non-detects). A separate limit of detection (LOD) based on signal-to-noise criteria or blank variability was not determined in this study. Method repeatability was evaluated through multiple analytical runs conducted on different days, showing consistent retention times, peak shapes, and analyte-to-internal standard response ratios for both native compounds and their isotopically labeled analogs. Matrix-related interferences were minimized through a

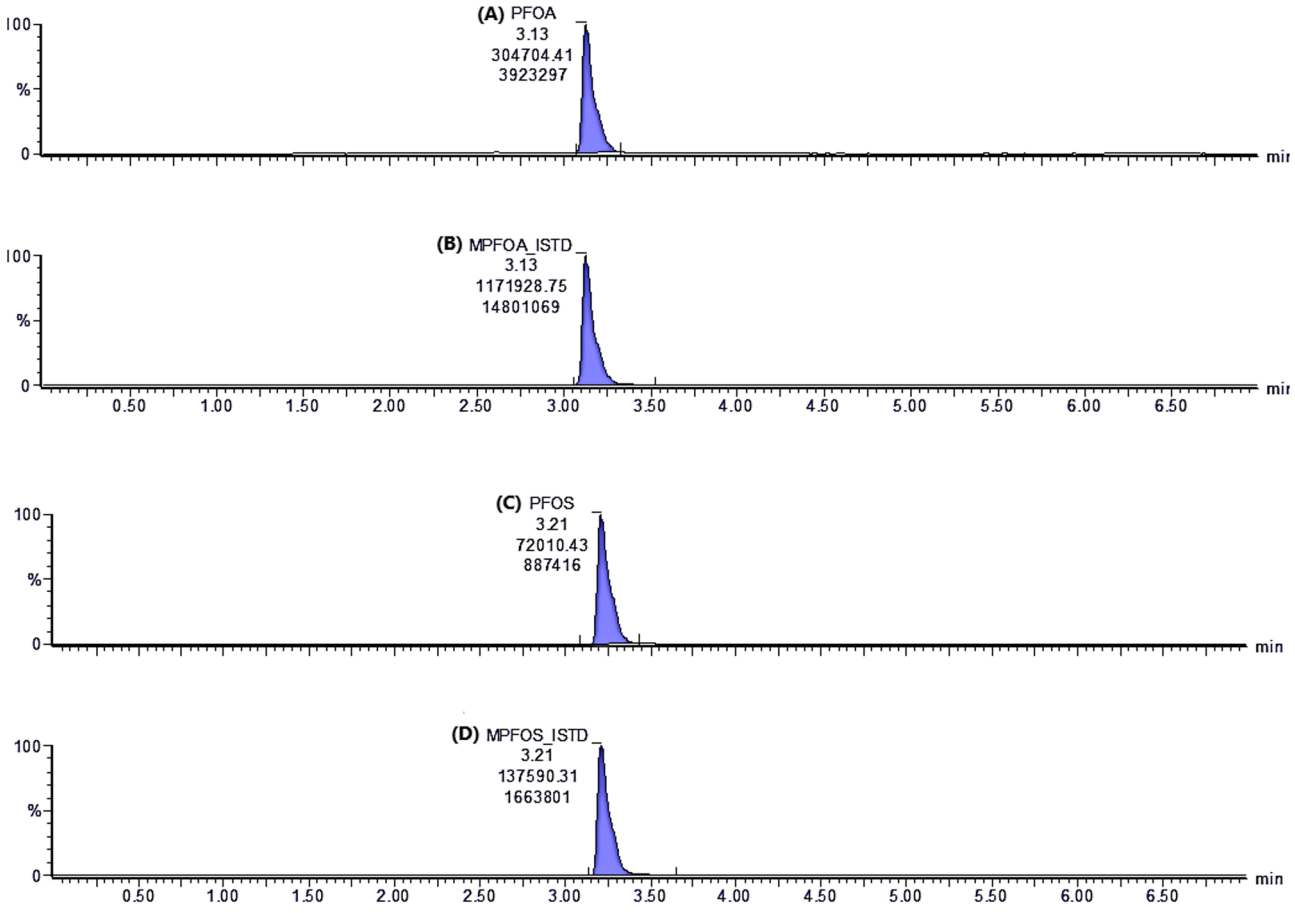


Fig. 1. Representative MRM chromatograms of target PFAS compounds and their isotopically labeled internal standards in human breast milk extracts obtained by LC-MS/MS. (**A**) **PFOA** native compound eluting at 3.13 min; (**B**) **MPFOA** isotopically labeled internal standard eluting at 3.13 min; (**C**) **PFOS** native compound eluting at 3.21 min; and (**D**) **MPFOS** isotopically labeled internal standard eluting at 3.21 min. All analytes were acquired using electrospray ionization in negative mode and quantified using MRM transitions as detailed in Table 3.

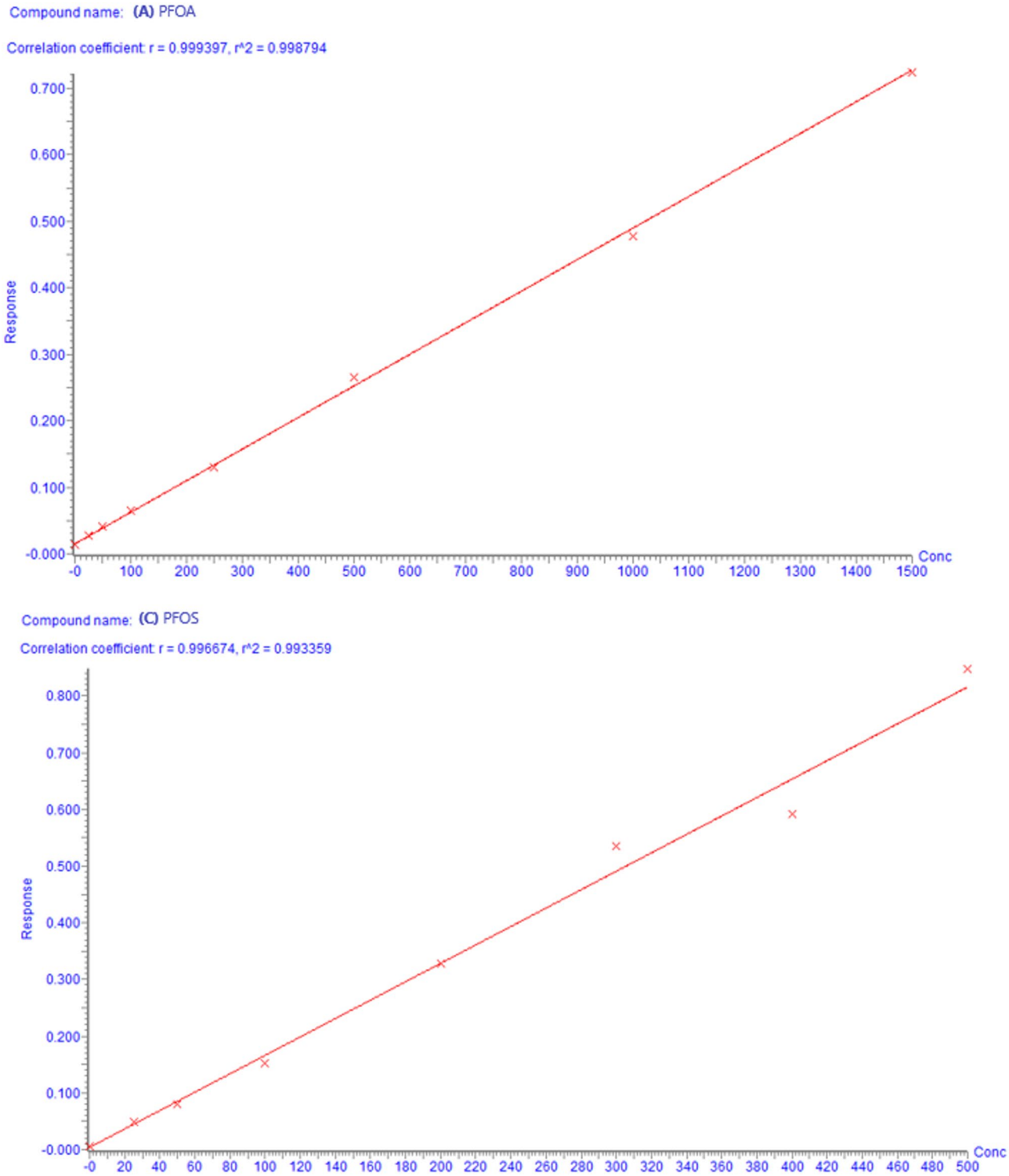


Fig. 2. Calibration curves for PFOA (**A**) and PFOS (**B**) using isotopically labeled internal standards obtained by LC-MS/MS. *Note*: Each calibration was constructed using seven concentration levels (PFOA: 25–1500 ng/L; PFOS: 25–500 ng/L). Both analytes demonstrated excellent linearity with correlation coefficients of $R^2 = 0.9987$ for PFOA and $R^2 = 0.9933$ for PFOS.

Analyte	Mean	Median	Min	Max
PFOA	< LLOQ	< LLOQ	< LLOQ	73.35
PFOS	< LLOQ	< LLOQ	< LLOQ	85.2

Table 4. Summary statistics for PFOA and PFOS concentrations in breast milk samples from 25 participants (ng/L). LLOQ = 25 ng/L.

(Reference, Year of Sample Collection)	Origin	*n*	Range			Median
			PFOA	PFOS	PFOA	PFOS
Al-Sheyab et al.[35]	Jordan	79	24–1120	< 10–178	82.5	50
Current study	Saudi Arabia	25	< **25–73.3**	< **25–85.2**	**< 25**	**< 25**
Fiedler and Sadia[41]	Africa	14	6.20–18.1	0-21.9	12.5	10.3
Yao et al.[29]	China	1151	12–668.9	2.6–87.7	336	49.7
Lee et al.[36]	South Korea	128	< 10–657	14.8–380	40.1	47.8
Beser et al.[40]	Spain	20	n.d.-180	n.d–78	138	69
Cariou et al.[38]	France	100	< 40–308	< 30–376	< 40	< 30
Abdallah et al.[37]	Ireland	16	16–350	< 20–120	100	20
Rawn et al.[39]	Canada	664	< 5-284	< 10–142	34.2	30.2

Table 5. Levels of PFOA and PFOS in human milk from published studies (ng/L).

combination of protein precipitation, solid-phase extraction, chromatographic separation, and internal standard normalization.

Calculation of PFOA and PFOS estimated daily intake via breast milk

The estimated daily intake (EDI) of PFOA and PFOS via breast milk was calculated using established exposure assessment methods (e.g., Xiao et al. 2023) and expressed as nanograms per kilogram body weight per day (ng/kg-bw/day), according to the following equation:

$$EDIs = \frac{C_{BM} \times V_{BM}}{BW},$$

where C_{BM} is the concentration of PFAS measured in breast milk (ng/L), V_{BM} is daily breast milk intake volume (L/day), and BW is the infant body weight (kg)[29]. For analytical consistency, the median concentration derived from four independent analytical runs was used to represent C_{BM} for each participant. Concentrations below the lower limit of quantification (LLOQ = 25 ng/L) were classified as non-detects (< LLOQ) and excluded from quantitative exposure calculations. Daily breast milk intake VBM was estimated as an absolute intake volume of 292.5 mL/day for infants ≤ 1 month of age. This value was derived by multiplying an age-specific intake rate of 117 mL/kg body weight per day—obtained from a global systematic review and meta-analysis of breast milk intake in infants (Rios-Leyvraz & Yao, 2023; PMID: 36763610)—by a reference infant body weight of 2.5 kg, representative of the average birth weight of Saudi neonates as reported by the Saudi Ministry of Health[30,31].The reference body weight was used solely to derive the absolute daily milk intake volume while exposure normalization per kilogram body weight was applied only once within the EDI equation, ensuring methodological consistency and avoiding double body-weight normalization. The estimated daily intake (EDI) of PFOA and PFOS via breast milk was calculated for all participants and expressed as ng/kg-bw/day.

Results

Concentration of PFOA and PFOS in breast milk

Summary statistics for PFOA and PFOS concentrations measured in breast milk are presented in Table 4. Because the majority of samples were below the analytical lower limit of quantification (LLOQ = 25 ng/L), the mean, median, and minimum values for both analytes fell below the quantification threshold. Quantifiable concentrations above the LLOQ were observed in four participants (16%), with maximum concentrations of 73.35 ng/L for PFOA and 85.2 ng/L for PFOS.

Individual breast milk concentrations of PFOA and PFOS for the 25 study participants are presented in Fig. 3.Quantifiable concentrations above the lower limit of quantification (LLOQ = 25 ng/L) were observed in four participants only. In these samples, both PFOA and PFOS were detected at concentrations exceeding the LLOQ, whereas all remaining samples were classified as non-detects (< LLOQ). PFOA concentrations ranged from LLOQ to 73.3 ng/L while PFOS concentrations ranged from LLOQ to 85.2 ng/L.

PFOA and PFOS estimated daily intake via breast milk

The distribution of estimated daily intake for both compounds is presented as a box-and-whisker plot in Fig. 4. Concentrations < LLOQ were operationally treated as non-detects and excluded from quantitative exposure

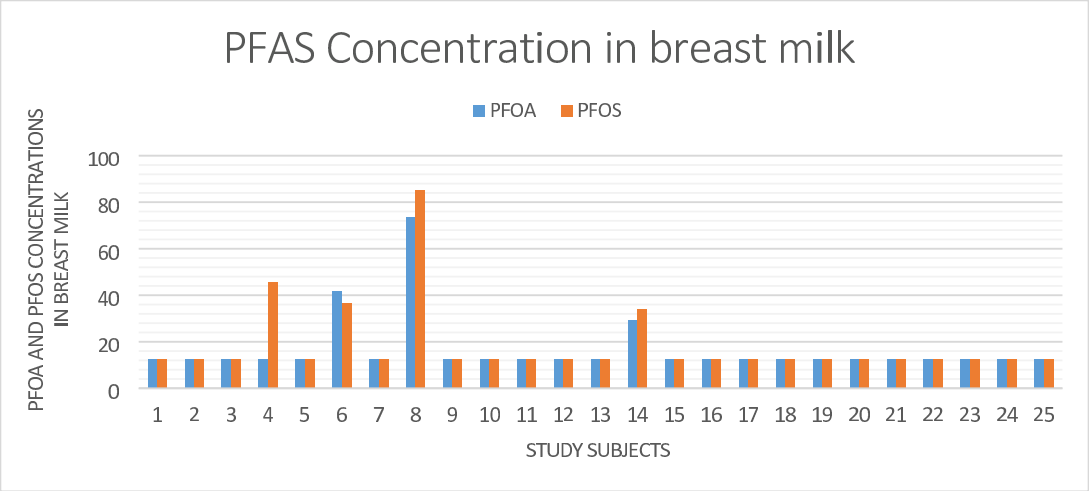


Fig. 3. PFAS concentration (PFOA and PFOS) from breast milk (ng/L)of 25 Saudi women. *Note* For visualization purposes only, concentrations below the LLOQ were substituted with half the LLOQ value (12.5 ng/L) to allow plotting across all participants. This substitution was applied for graphical display only and did not influence quantitative reporting or statistical analyses. The Y-axis represents the median concentrations of PFOA and PFOS in breast milk expressed in (ng/L), obtained after analytical replication across three separate days. The X-axis denotes individual study participants ($n = 25$).

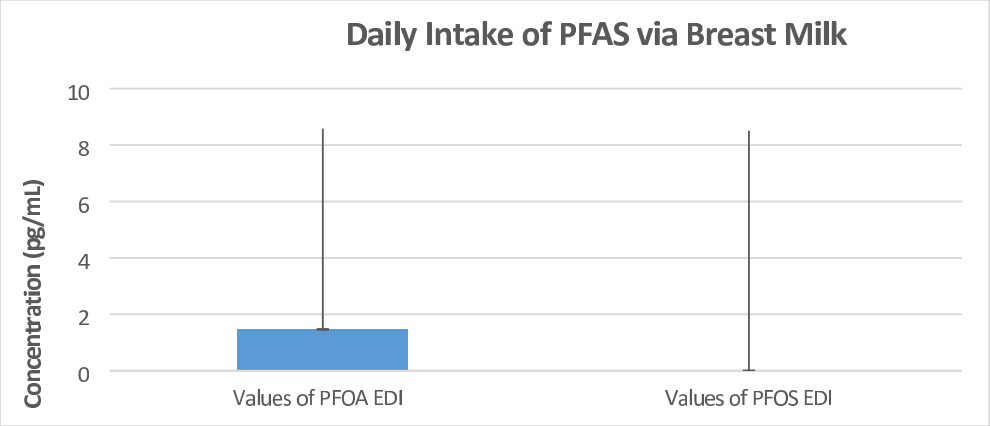


Fig. 4. Estimated daily intake of PFOA and PFOS via breast milk (ng/kg bw/day). *Note*: Both PFOA and PFOS EDI distributions were highly censored, with nearly all values clustered at the minimum intake level (1.46 ng/kg-bw/day). As a result, box plot visualization showed negligible interquartile range and narrow whiskers, reflecting limited variability across the samples. Only one subject in each compound exhibited a quantifiable EDI value above the censored baseline. Box plot visualization revealed a visible IQR for PFOA due to the presence of intermediate intake values between the censored baseline and the upper quantifiable value. In contrast, PFOS values were almost entirely censored at the minimum intake level, resulting in a collapsed IQR and no box formation, with only the upper whisker representing the single quantifiable observation.

calculations. For visualization purposes only, non-detect values were plotted as LLOQ/2, a standard convention in censored environmental datasets. Given the low proportion of quantifiable values, substitution approaches (e.g., LLOQ/2 or LOD/$\sqrt{2}$) had negligible influence on distributional parameters and did not alter study interpretation. This approach aligns with PFAS biomonitoring conventions applied in previous studies.

Discussion

The present study quantified PFOA and PFOS in milk from lactating women in Riyadh, Saudi Arabia. Although most samples exhibited concentrations below the analytical LLOQ, quantifiable values were detected in a subset of participants, confirming the presence of PFAS exposure. The distribution of measured concentrations was markedly right-skewed, suggesting that exposure is not uniform across the population and may be influenced by individual-specific determinants such as dietary habits, consumer product use, lifestyle, and environmental sources. While such determinants were not assessed in this exploratory study, the observed pattern underscores

the need for future investigations aimed at identifying exposure predictors in the Saudi context. Previous environmental investigations in Saudi Arabia—including studies of coastal waters and marine organisms—have detected PFAS contamination, suggesting plausible environmental and dietary exposure pathways[32,33]. Wastewater effluent discharge and seafood consumption have been proposed as potential contributors in the Saudi context[32,33]. In addition to environmental evidence, human biomonitoring data from Saudi Arabia further support the presence of PFAS exposure in the population. A recent study conducted in Jeddah reported measurable serum concentrations of PFOA and PFOS compounds among adults and explored their association with osteoporosis[34]. While the biological matrix, study population, and health endpoints differ from the present investigation, these findings provide complementary evidence of PFAS occurrence in human biological samples within Saudi Arabia. Together, environmental and serum-based studies support the plausibility of background PFAS exposure in the region and underscore the relevance of assessing early-life exposure biomarkers such as breast milk.

To contextualize these findings, breast milk concentrations of PFOA and PFOS were compared against studies conducted in various regions worldwide[21]. Only studies published after 2015 were considered for direct comparison due to major international regulatory actions initiated during this period—most notably the U.S. EPA 2010/2015 PFOA Stewardship Program, which committed participating industries to phase out long-chain PFAS emissions and product content by 2015[9]. Post-2015 datasets therefore provide a more accurate reflection of contemporary human exposure. The studies included for comparison are summarized in Table 5.Across these post-2015 datasets, the highest upper-range concentrations of PFOA were observed in Jordan (up to 1120 ng/L), followed by China (up to 668.9 ng/L) and South Korea (up to 657 ng/L), whereas Ireland, France, Canada, and Spain reported intermediate ranges[29,35–40]. For PFOS, the highest upper-bound levels were reported in South Korea (up to 380 ng/L), followed closely by France (up to 376 ng/L)[36,38]. In terms of central tendency, China exhibited the highest reported median PFOA level (336 ng/L), whereas median PFOS values were generally comparable in China, South Korea, and Jordan (approximately 47–50 ng/L)[29,35,36]. In contrast, PFAS concentrations reported from African cohorts were markedly lower, approximating those observed in the current Saudi study, where median values for both PFOA and PFOS remained below the analytical LLOQ[41]. These international discrepancies likely reflect multiple interacting factors, including differences in PFAS manufacturing and use, industrial phase-out timelines, environmental emissions, wastewater and sludge management practices, and the continued use of PFAS-containing consumer products. Methodological differences—including sampling year, milk collection period, analytical technique, LLOQ thresholds, and biological matrices—may further contribute to interstudy variation. Demographic characteristics such as parity, diet, and exposure pathways may also influence breast milk PFAS concentrations. Such contextual and methodological heterogeneity must be considered when interpreting international comparisons.

To further assess infant exposure during early postnatal life, Estimated Daily Intake (EDI) values for both PFOA and PFOS were calculated from breast milk concentrations. When evaluated against the United States Environmental Protection Agency (U.S. EPA) oral reference dose (RfD) of 20 ng/kg-bw/day, all samples fell well below the U.S. benchmark. In contrast, when assessed against the more conservative European Food Safety Authority (EFSA) reference values—0.857 ng/kg-bw/day for PFOA and 1.857 ng/kg-bw/day for PFOS—a subset of samples exceeded these thresholds[42,43]. The divergence between U.S. EPA and EFSA benchmarks illustrates the considerable regulatory variability in defining acceptable PFAS exposure levels during infancy and underscores ongoing debate regarding potential health risks associated with low-dose chronic exposure during critical developmental windows. However, exceedance of EFSA reference values does not imply clinical toxicity, these findings highlight early infancy as a physiologically vulnerable period, during which breastfed infants receive higher exposure doses per kilogram body weight than older children and adults. EFSA reference values are derived from developmental and immunotoxicity endpoints—such as reduced vaccine antibody response—which are particularly relevant during early postnatal immune maturation. Importantly, the present study did not assess health outcomes, and the EDI values should therefore be interpreted strictly as exposure metrics rather than indicators of adverse effects. This pattern justifies continued biomonitoring and source tracking in the region to better characterize exposure determinants and potential public health implications.

Conclusion

This study provides the first biomonitoring evidence of PFOA and PFOS in human breast milk in Saudi Arabia. Although most concentrations were below the analytical quantification limit, measurable levels in a subset of participants confirm ongoing PFAS exposure. Estimated daily intake values were below the U.S. EPA reference dose but exceeded EFSA benchmark values in a minority of samples, illustrating current regulatory uncertainty regarding acceptable early-life exposure. Compared with post-2015 international datasets, PFAS concentrations in milk from Saudi Arabia were substantially lower than those reported in East Asian and European cohorts and comparable to African datasets.

Study strengths

This study demonstrates multiple methodological strengths that enhance the relevance and rigor of its findings. It represents the first biomonitoring investigation of PFAS in human breast milk in Saudi Arabia, addressing a previously uncharacterized geographic data gap within the global PFAS exposure literature. Analytical quantification was performed using isotope-dilution LC-MS/MS, the reference standard for PFAS measurement, thereby improving both analytical specificity and quantitative accuracy. Estimated daily intake calculations incorporated age-specific breast milk consumption rates, which increase the relevance for early-life exposure assessment. In addition, international comparisons were restricted to datasets collected during post-regulatory

periods, reducing historical variability associated with legacy PFAS production and improving interpretability within contemporary exposure contexts.

Study limitations

Despite these strengths, several limitations should be considered when interpreting the results. The sample size was modest ($n = 25$), limiting statistical power and constraining the ability to examine exposure determinants or conduct stratified analyses. As the study was designed as an exploratory biomonitoring effort rather than a hypothesis-driven epidemiological study, no a priori power calculation was performed. Larger population-based investigations are therefore warranted to validate these findings and to more accurately characterize population variability. Although maternal age and parity were recorded, other potential determinants of PFAS exposure—such as dietary patterns, drinking water source, maternal BMI, consumer product use, and proximity to industrial sources—were not collected, precluding attribution of exposure differences to specific behavioral or environmental pathways. Furthermore, most PFOA and PFOS measurements fell below the analytical LLOQ, which resulted in right-censoring of the dataset. While this reflects low exposure levels in the population, it limits the precision of statistical summaries and data visualization; future analyses employing approaches suitable for censored data (e.g., Tobit models or multiple imputation) may improve interpretability. The study also employed a cross-sectional design limited to early postpartum milk, providing only a narrow temporal snapshot. PFAS concentrations in breast milk may vary by lactation stage due to physiological mobilization, cumulative infant transfer, or metabolic changes, suggesting that longitudinal sampling would offer a more comprehensive understanding of temporal dynamics. Infant exposure estimation relied on standardized assumptions regarding milk consumption and body weight, which, although widely used in exposure science, may not reflect individual-level variation. In addition, data on direct sources of exposure—such as drinking water quality, dietary intake patterns, or consumer product use—were not collected. This absence of source attribution data limits the ability to contextualize measured concentrations and to identify dominant exposure pathways in the Saudi population. Lastly, the study quantified exposure but did not assess clinical outcomes, toxicological biomarkers, or infant health. Accordingly, the findings should be interpreted within the context of exposure characterization rather than health risk evaluation.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

EA, II and EG conception, preparation of samples, writing and reviewing of manuscript EA, WA and AA sample preparation, and data interpretation MA, EA and SA sample collection from hospitalII, RA, MA and BA manuscript review and correction. All authors finalized manuscript with inputs.

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Declarations

Ethics approval and consent to participate

This study was approved by KAIMRC/MNGHA Institutional Review Board (IRB) and written consent was obtained from subjects. Blank forms can be provided upon request.

Competing interests

The authors declare no competing interests.

Ethics declaration

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Disclosure

All the authors declare that there is no conflict of interest toward this publication.

Additional information

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